

Einstein's General Theory of Relativity

Lecture 5: Tensor Calculus, Connections, and Geodesics

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Chapter 1

Tensor Calculus, Connections, and Geodesics

Overview. Tensor algebra tells us how components transform. Tensor calculus must do something harder: compare fields at neighboring points without confusing a change of the field with a change of the coordinate basis. We will see the ordinary derivative fail, repair it with a connection, determine the Levi-Civita connection from the metric, and use it to define geodesics.

1.1 Why Tensor Equations Matter

Compatible tensors may be added and subtracted, but their index structures must match. A covector cannot be added directly to a vector, and a two-index tensor cannot be added to a three-index tensor. This is more than bookkeeping: each index records one factor in the coordinate transformation law.

The reason for organizing physics this way is that geometry exists before coordinates are chosen. A surface may be curved, a mountain may rise from it, and two points may have a definite separation even if no latitude or longitude has been assigned. Coordinates become indispensable when we want numbers and calculations. Tensors let us introduce those numbers without making the truth of an equation depend on the labels.

For example, a covector transforms according to

$$V_m^{(y)} = \frac{\partial x^r}{\partial y^m} V_r^{(x)}, \quad (1.1)$$

and a mixed tensor according to

$$(T^m_n)_{(y)} = \frac{\partial y^m}{\partial x^r} \frac{\partial x^s}{\partial y^n} (T^r_s)_{(x)}. \quad (1.2)$$

If every component on the right vanishes, every component on the left vanishes. Thus a tensor equation such as

$$T^m_n = 0 \quad (1.3)$$

is either true in every coordinate system or false in every coordinate system. The same is true of an equality between tensors with matching indices.

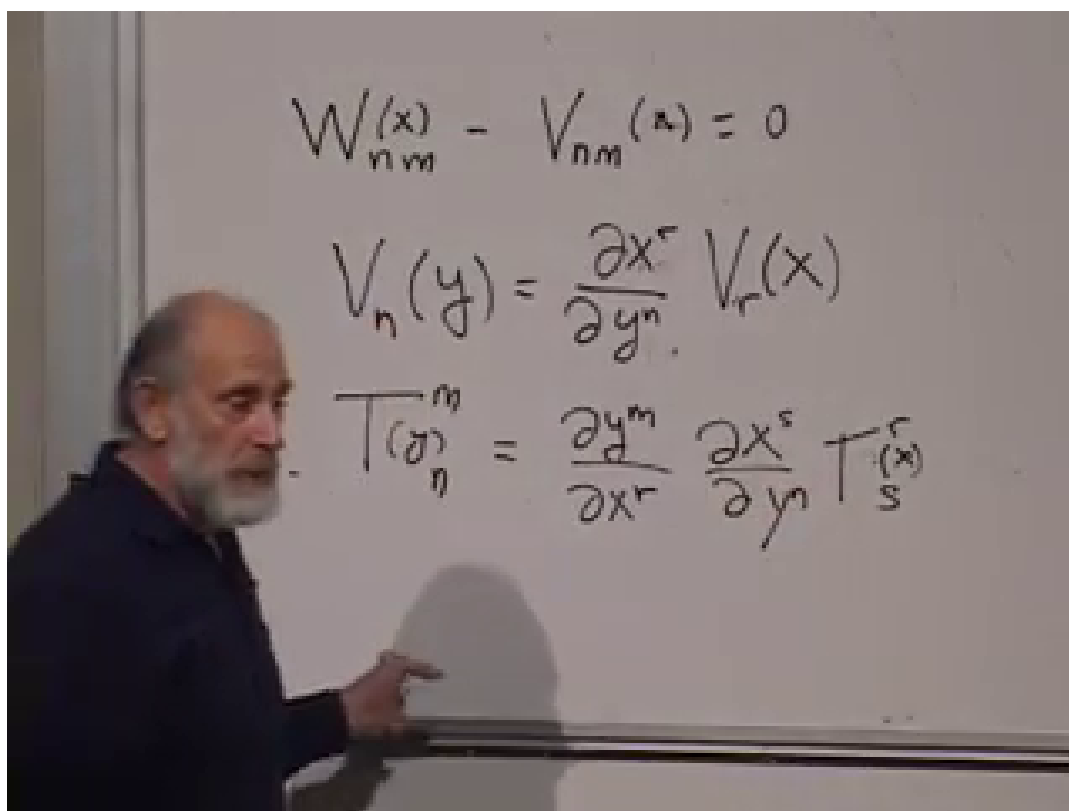


Figure 1.1: Transformation laws make the zero tensor coordinate independent. The board displays covariant and mixed examples before the lecture turns from tensor algebra to tensor calculus.

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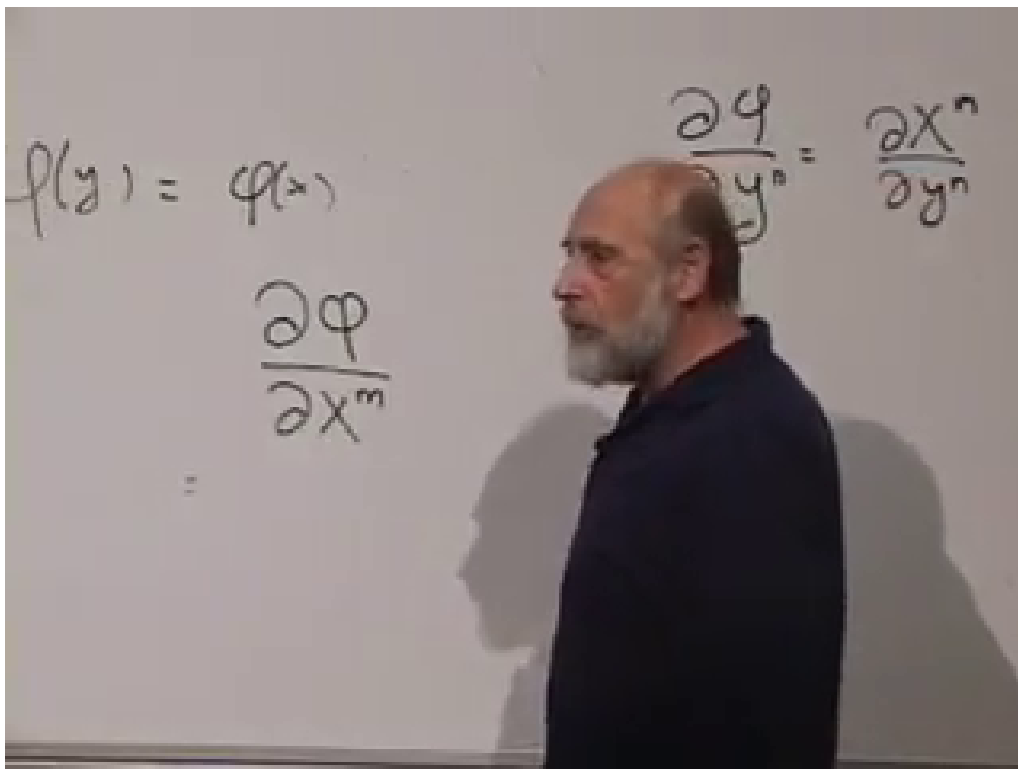


Figure 1.2: The scalar chain rule. The gradient acquires one inverse Jacobian and hence one lower index.

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General relativity needs differential equations, not only algebraic relations. We therefore need a derivative of a tensor that is again a tensor. The ordinary partial derivative succeeds for scalars and fails for vectors; understanding that failure is the central problem of this lecture.

1.2 The Scalar Control Case

A scalar field assigns one number to each point. Under a coordinate change the number is unchanged:

$$\phi^{(y)}(y) = \phi^{(x)}(x). \quad (1.4)$$

The chain rule gives

$$\frac{\partial \phi}{\partial y^n} = \frac{\partial x^m}{\partial y^n} \frac{\partial \phi}{\partial x^m}. \quad (1.5)$$

This is precisely the transformation law for a covector. The gradient of a scalar is therefore a tensor.

The physical content is equally simple. If the temperature is constant throughout a room, all of its derivatives vanish in Cartesian coordinates. Changing to polar, spherical, or any other smooth coordinates cannot make the temperature vary. Equation (1.5) guarantees exactly that.

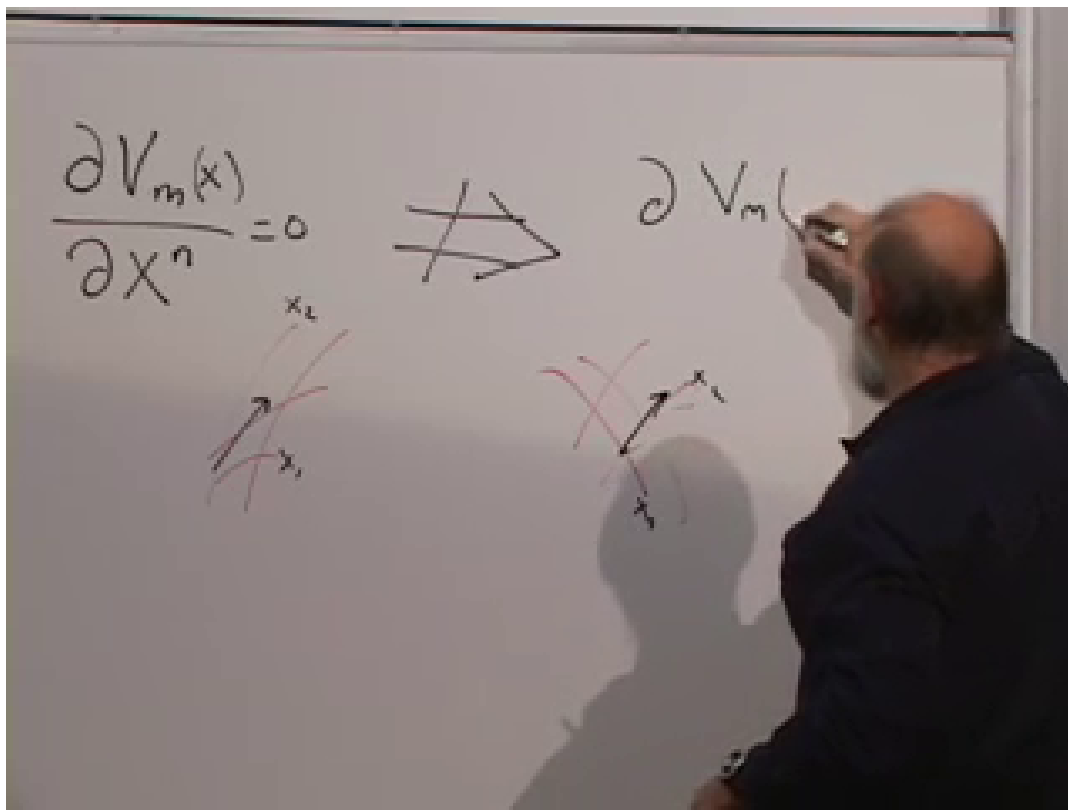


Figure 1.3: One geometrically constant vector field viewed through straight and curving coordinate grids. The components change because the local axes change.

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1.3 A Constant Vector in a Moving Frame

Now consider a vector field in an ordinary flat plane. Suppose every arrow has the same magnitude and points in the same geometric direction. In Cartesian coordinates its components are constant:

$$\frac{\partial V_m^{(x)}}{\partial x^n} = 0. \quad (1.6)$$

Lay a curvilinear grid over the same plane. The field has not changed, but the coordinate directions turn from point to point. Components are projections onto those local directions, so the y -components generally change:

$$\frac{\partial V_m^{(y)}}{\partial y^n} \neq 0. \quad (1.7)$$

Question from the audience. Does saying that the vector has one fixed direction already assume some third reference frame common to both coordinate systems? ^a

Response. The statement is geometric. In the flat plane we may compare arrows directly, independently of either curvilinear description. Cartesian coordinates make that comparison especially transparent, but the vector field does not acquire its direction from the coordinates. The coordinates only supply components.

^aLecture timestamp: 00:18:28–00:20:17.

This example identifies the difficulty. The ordinary derivative of vector components measures two effects at once: variation of the vector and variation of the basis. A tensorial derivative must remove the second effect.

1.4 The Ordinary Derivative Fails the Tensor Test

Let the x -coordinates be Cartesian for the moment and define

$$T_{mn}^{(x)} \stackrel{?}{=} \frac{\partial V_m^{(x)}}{\partial x^n}. \quad (1.8)$$

If this were a genuine rank-two covariant tensor, its y -components would have to be

$$\begin{aligned} T_{mn}^{(y)} &= \frac{\partial x^r}{\partial y^m} \frac{\partial x^s}{\partial y^n} T_{rs}^{(x)} \\ &= \frac{\partial x^r}{\partial y^m} \frac{\partial x^s}{\partial y^n} \frac{\partial V_r^{(x)}}{\partial x^s} \\ &= \frac{\partial x^r}{\partial y^m} \frac{\partial V_r^{(x)}}{\partial y^n}. \end{aligned} \quad (1.9)$$

Now calculate the ordinary derivative of the transformed covector directly. From Eq. (1.1),

$$\begin{aligned} \frac{\partial V_m^{(y)}}{\partial y^n} &= \frac{\partial}{\partial y^n} \left(\frac{\partial x^r}{\partial y^m} V_r^{(x)} \right) \\ &= \frac{\partial x^r}{\partial y^m} \frac{\partial V_r^{(x)}}{\partial y^n} + \frac{\partial^2 x^r}{\partial y^n \partial y^m} V_r^{(x)}. \end{aligned} \quad (1.10)$$

The first term is Eq. (1.9). The second term is new. It comes from differentiating the Jacobian that describes the moving coordinate basis.

Question from the audience. Has the derivative of the Jacobian already been counted inside the chain-rule term? ^a

Response. No. The chain-rule term differentiates the vector components while holding the Jacobian factor outside. Directly differentiating their product also differentiates the Jacobian. That second derivative of the coordinate map is the missing term, and it is exactly why the two expressions are unequal.

^aLecture timestamp: 00:31:34–00:32:29.

Thus

$$\frac{\partial V_m^{(y)}}{\partial y^n} \neq \frac{\partial x^r}{\partial y^m} \frac{\partial x^s}{\partial y^n} \frac{\partial V_r^{(x)}}{\partial x^s} \quad (1.11)$$

for a general coordinate transformation.

$$T_{mn}(x) = \frac{\partial V_m(x)}{\partial x^n}$$

$$T_{mn}(y) = \frac{\partial x^r}{\partial y^m} \frac{\partial x^s}{\partial y^n} T_{rs}(x) = \frac{\partial x^r}{\partial y^m} \frac{\partial V_m(x)}{\partial x^n}$$

?

Figure 1.4: The tentative tensor law for $T_{mn} = \partial_n V_m$. The question mark is essential: this is the hypothesis being tested, not yet an identity.

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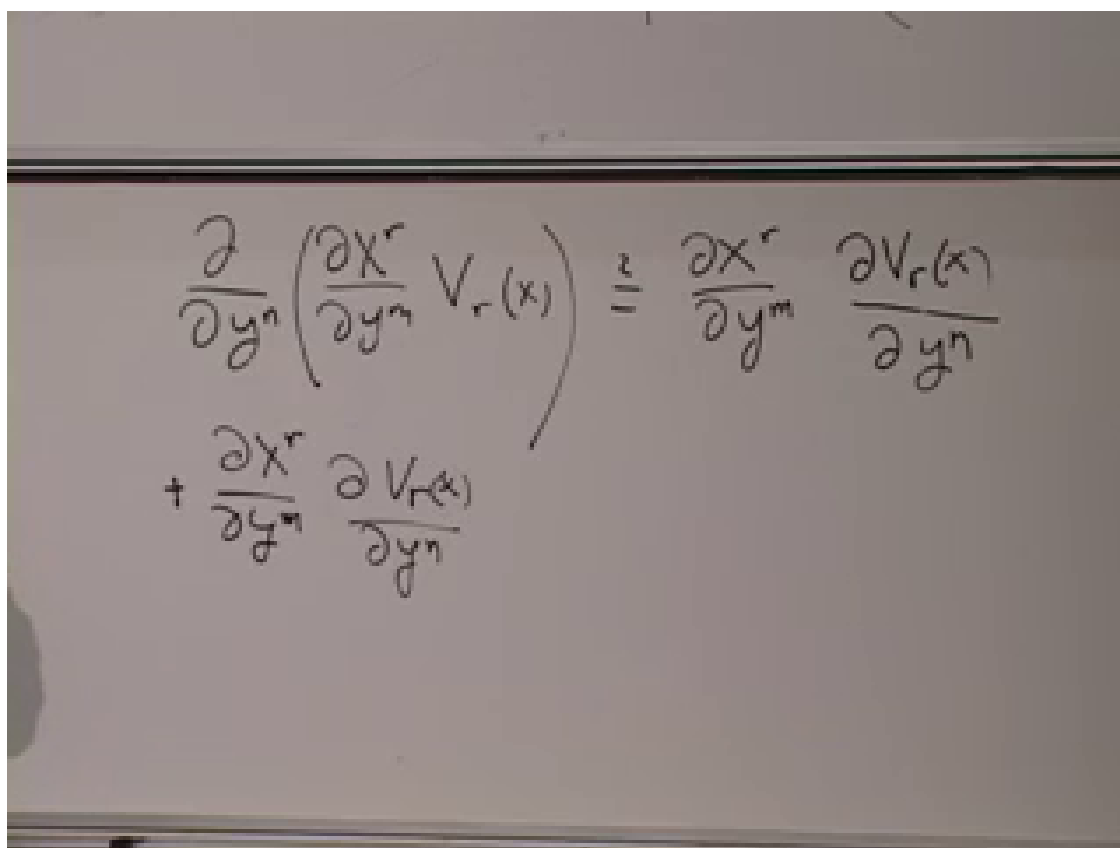

$$\frac{\partial}{\partial y^n} \left(\frac{\partial x^r}{\partial y^m} V_r(x) \right) = \frac{\partial x^r}{\partial y^m} \frac{\partial V_r(x)}{\partial y^n} + \frac{\partial x^r}{\partial y^n} \frac{\partial V_r(x)}{\partial y^m}$$

Figure 1.5: The product rule exposes the extra second-derivative term. This term vanishes for a linear coordinate transformation but not for a general curvilinear one.

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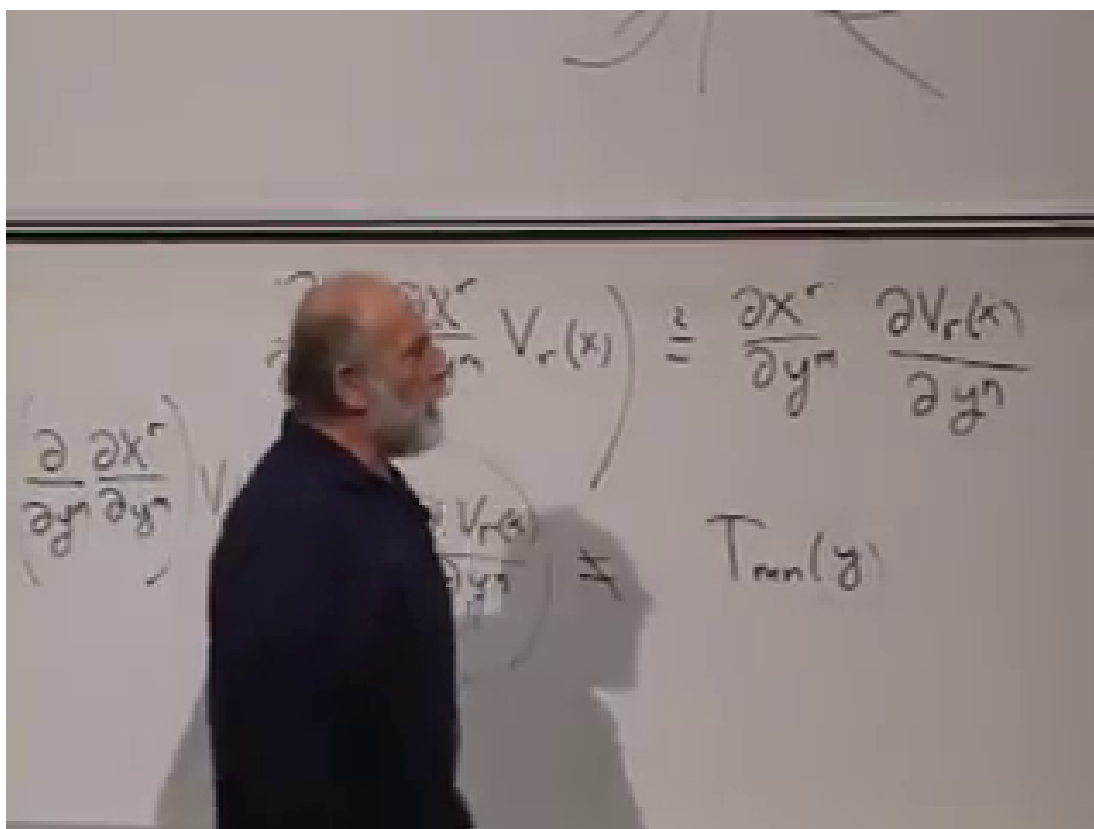


Figure 1.6: The completed comparison: the extra product-rule term prevents the ordinary component derivative from being a tensor.

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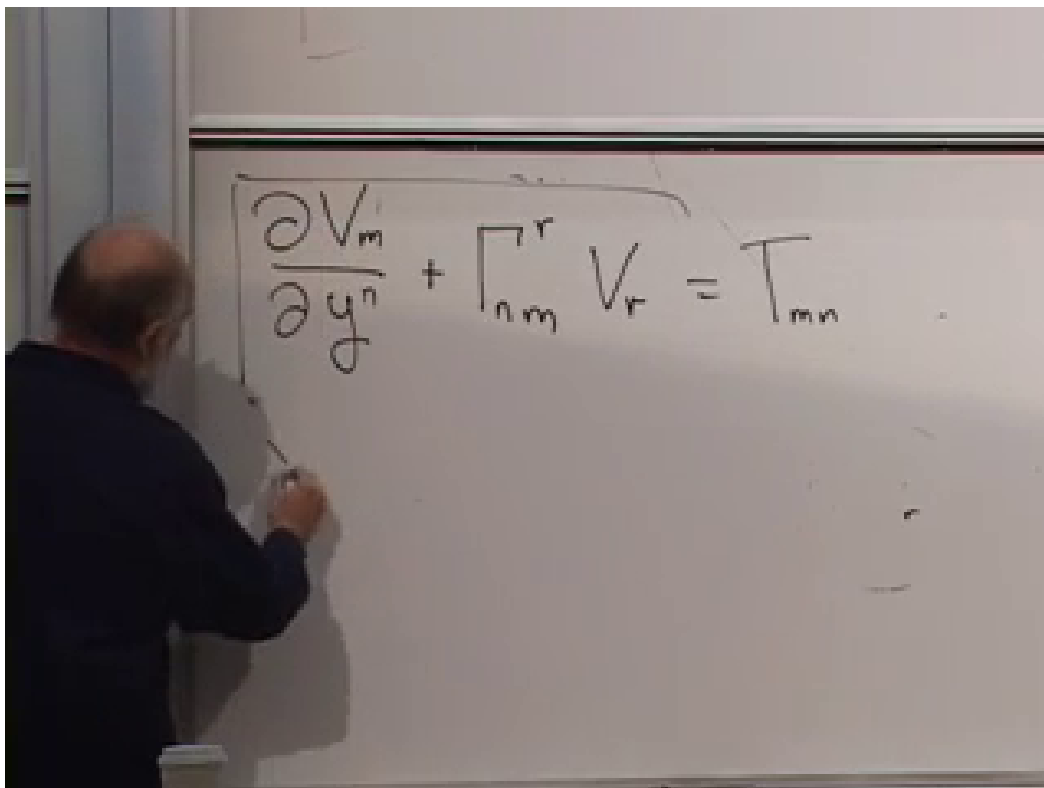


Figure 1.7: The first corrected derivative on the board. To keep the later Christoffel formula conventional, we denote the temporary plus-sign coefficient here by $C^r_{nm} = -\Gamma^r_{nm}$.

Lecture frame: 00:40:30.

Question from the audience. Can this failure be seen in a physical field, such as an electric field? ^a

Response. Yes. A uniform electric field has constant Cartesian components. In curvilinear coordinates the same field generally has position-dependent components because the coordinate axes rotate. The ordinary derivatives then report a change even though the physical field is uniform.

^aLecture timestamp: 00:33:58–00:34:47.

For a linear transformation the Jacobian is constant, so the extra term vanishes. That special case explains why ordinary component derivatives behave harmlessly under rotations and other rigid linear changes, while general coordinate transformations reveal the problem.

1.5 From the Extra Term to the Connection

The failure suggests its own cure. We seek a correction linear in the vector,

$$\partial_n V_m + C^r_{nm} V_r, \tag{1.12}$$

whose inhomogeneous transformation cancels the derivative of the Jacobian. The board initially denotes this temporary coefficient by Γ and writes it with a plus sign for a lower index.

We now adopt the standard Levi–Civita sign convention by defining

$$C^r_{nm} = -\Gamma^r_{nm}. \quad (1.13)$$

The covariant derivative of a covector is therefore

$$\boxed{\nabla_n V_m = \partial_n V_m - \Gamma^r_{nm} V_r.} \quad (1.14)$$

This one normalization resolves the apparent sign change between the early board equations and the conventional Christoffel formula written later. No geometry changes; only the name assigned to the temporary correction coefficient changes.

Question from the audience. Was the correction obtained only because the starting x -coordinates were Cartesian, or does a connection exist in a general geometry? ^a

Response. The Cartesian calculation is the cleanest way to expose the inhomogeneous term. A connection is defined in general coordinates and on curved spaces as well. For the metric geometry used here, the metric will determine the unique torsion-free, metric-compatible connection.

^aLecture timestamp: 00:42:09–00:42:53.

The word *covariant* in *covariant derivative* describes the coordinate-respecting operation. It does not mean that the differentiated object must carry a lower index.

1.5.1 One correction for each index

Each tensor index contributes its own connection term. With the standard sign convention, every lower index contributes a minus sign and every upper index contributes a plus sign. For a rank-two covariant tensor,

$$\boxed{\nabla_p T_{mn} = \partial_p T_{mn} - \Gamma^r_{pm} T_{rn} - \Gamma^r_{pn} T_{mr}.} \quad (1.15)$$

The lecture’s practical method is useful. Temporarily ignore all but one tensor index, apply the vector rule to that slot, and then repeat for every other slot. What must never change is the identity of the free indices.

Question from the audience. Which tensor index belongs in each correction term? ^a

Response. The connection acts on one slot at a time. Acting on the m -slot replaces m by the dummy index r , producing $\Gamma^r_{pm} T_{rn}$. Acting on the n -slot produces $\Gamma^r_{pn} T_{mr}$. The classroom corrections to the board are important: T_{rn} and T_{mr} cannot be interchanged without an independent symmetry.

^aLecture timestamp: 00:46:14–00:53:21.

Question from the audience. May a free index be swapped with a repeated index? ^a

Response. No. A repeated index is a dummy summation label and may be renamed consistently within its term. A free index identifies a component of the result and must remain free in the same position in every term. Indices may be interchanged only when a stated symmetry permits it.

^aLecture timestamp: 00:53:55–00:54:51.

For a vector with an upper index,

$$\boxed{\nabla_m V^n = \partial_m V^n + \Gamma^n_{mr} V^r.} \quad (1.16)$$

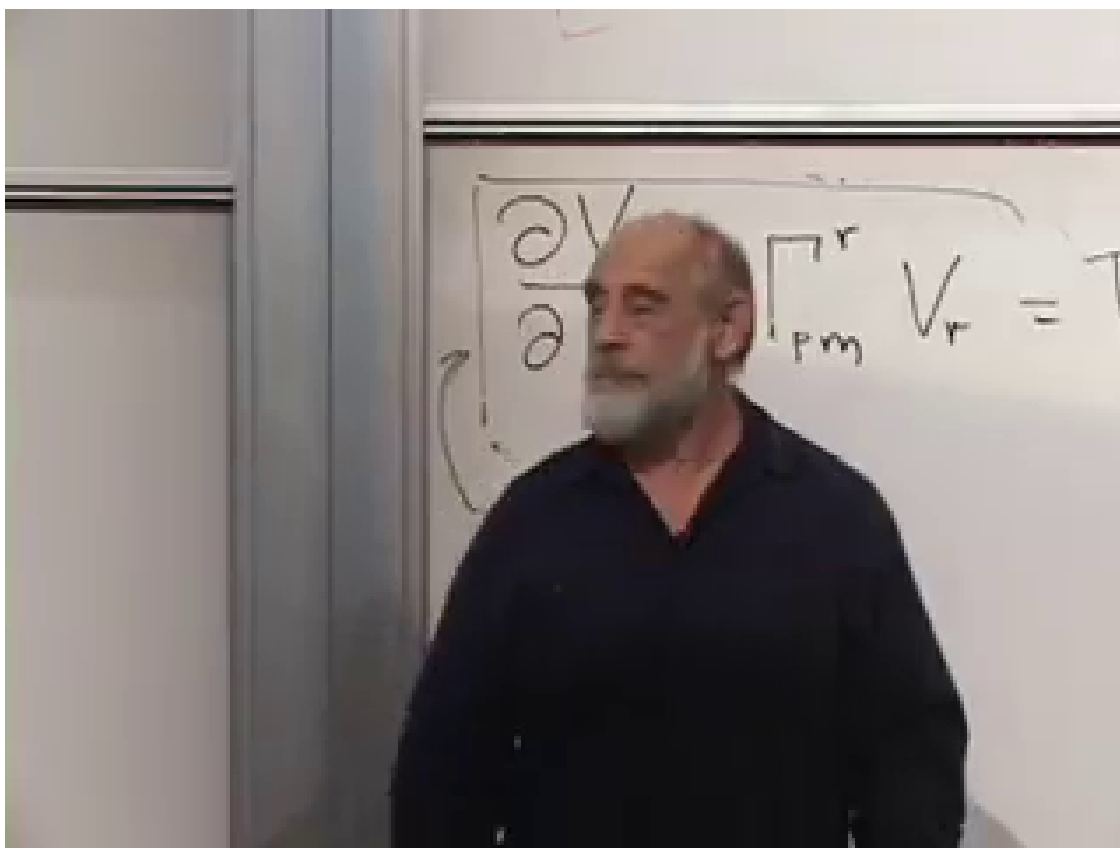


Figure 1.8: The board builds the rank-two rule one index at a time using the temporary $C = -\Gamma$ convention. In standard notation the two lower-index corrections carry minus signs, as in Eq. (1.15).

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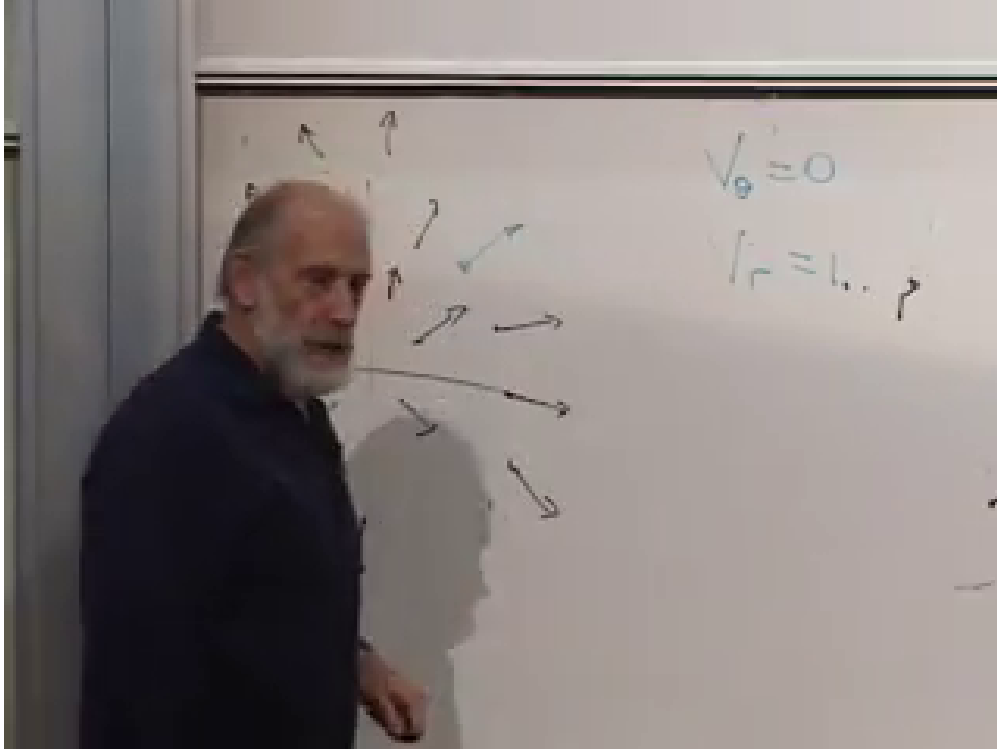


Figure 1.9: The radial field has constant polar components $V^r = 1$, $V^\theta = 0$, but its direction changes from point to point.

Lecture frame: 00:56:52.

A scalar has no tensor index on which the connection can act, so $\nabla_m \phi = \partial_m \phi$.

1.6 The Polar-Coordinate Test

The abstract rule becomes visible in polar coordinates. On the flat plane,

$$ds^2 = dr^2 + r^2 d\theta^2, \quad g_{rr} = 1, \quad g_{\theta\theta} = r^2. \quad (1.17)$$

Ignore the singular origin and consider the radial field

$$V^r = 1, \quad V^\theta = 0. \quad (1.18)$$

Its polar components are constant, yet the arrows point in different Cartesian directions around the origin. The field is not geometrically constant.

The nonzero polar Christoffel symbols are

$$\Gamma^r_{\theta\theta} = -r, \quad \Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = \frac{1}{r}. \quad (1.19)$$

Consequently,

$$\nabla_\theta V^\theta = \Gamma^\theta_{\theta r} V^r = \frac{1}{r} \neq 0. \quad (1.20)$$

The covariant derivative detects the turning of the field even though the ordinary derivatives of its polar components vanish.

Question from the audience. Does Cartesian simply mean that the coordinate lines look straight, as on a Mercator map? ^a

Response. No. Here Cartesian means that the flat-space metric has constant components $g_{mn} = \delta_{mn}$. A drawing may use straight grid lines while distorting the metric. Conversely, polar coordinates describe a flat plane even though their coordinate basis changes from point to point.

^aLecture timestamp: 00:59:32–01:00:30.

1.7 Metric Compatibility

The connection has so far been introduced as the correction needed for tensorial differentiation. We now determine it from the metric. In flat Cartesian coordinates the metric is constant, so its covariant derivative should vanish. Since $\nabla_p g_{mn}$ is a tensor, vanishing in those coordinates implies vanishing in every coordinate system:

$$\boxed{\nabla_p g_{mn} = 0.} \quad (1.21)$$

Using the standard convention,

$$0 = \partial_p g_{mn} - \Gamma^r_{pm} g_{rn} - \Gamma^r_{pn} g_{mr}. \quad (1.22)$$

Question from the audience. Why choose the covariant derivative of the metric to be zero? ^a

Response. In homogeneous flat space the metric is the same at every point in Cartesian coordinates, so its ordinary derivative vanishes. The covariant derivative is designed to preserve that geometric statement in arbitrary coordinates. More deeply, $\nabla g = 0$ says that parallel transport preserves lengths and angles.

^aLecture timestamp: 01:18:26–01:19:49.

Metric compatibility alone leaves the possibility of torsion. The connection used here is also taken to be torsion-free,

$$\Gamma^a_{bc} = \Gamma^a_{cb}. \quad (1.23)$$

These two conditions determine the Levi-Civita connection uniquely. Permuting the indices in Eq. (1.22), adding two equations, subtracting the third, and raising one index gives

$$\boxed{\Gamma^a_{bc} = \frac{1}{2} g^{ad} (\partial_b g_{dc} + \partial_c g_{db} - \partial_d g_{bc}).} \quad (1.24)$$

The Christoffel symbols are also called connection coefficients or an affine connection. They are built from the metric and its first derivatives. They are not themselves the components of a tensor.

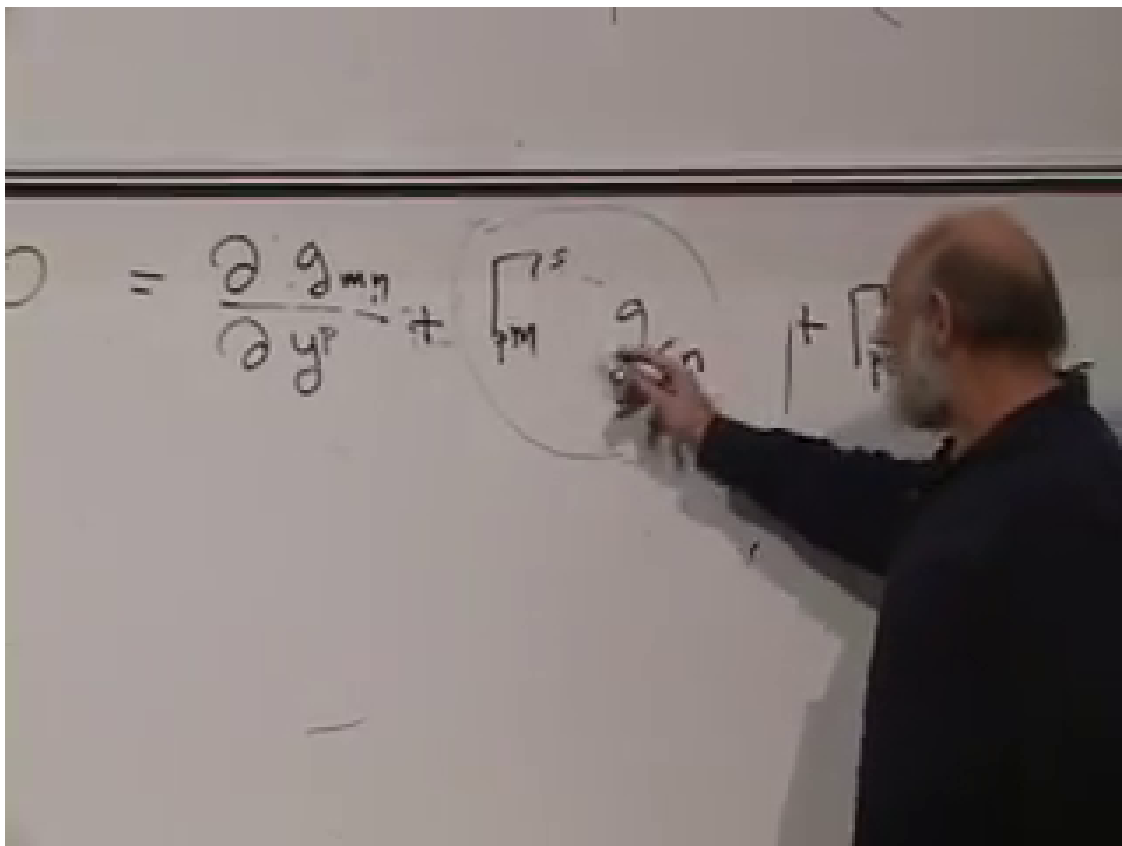


Figure 1.10: Metric compatibility expanded on the board in the temporary $C = -\Gamma$ notation. Converting to the standard Christoffel symbol yields Eq. (1.22).

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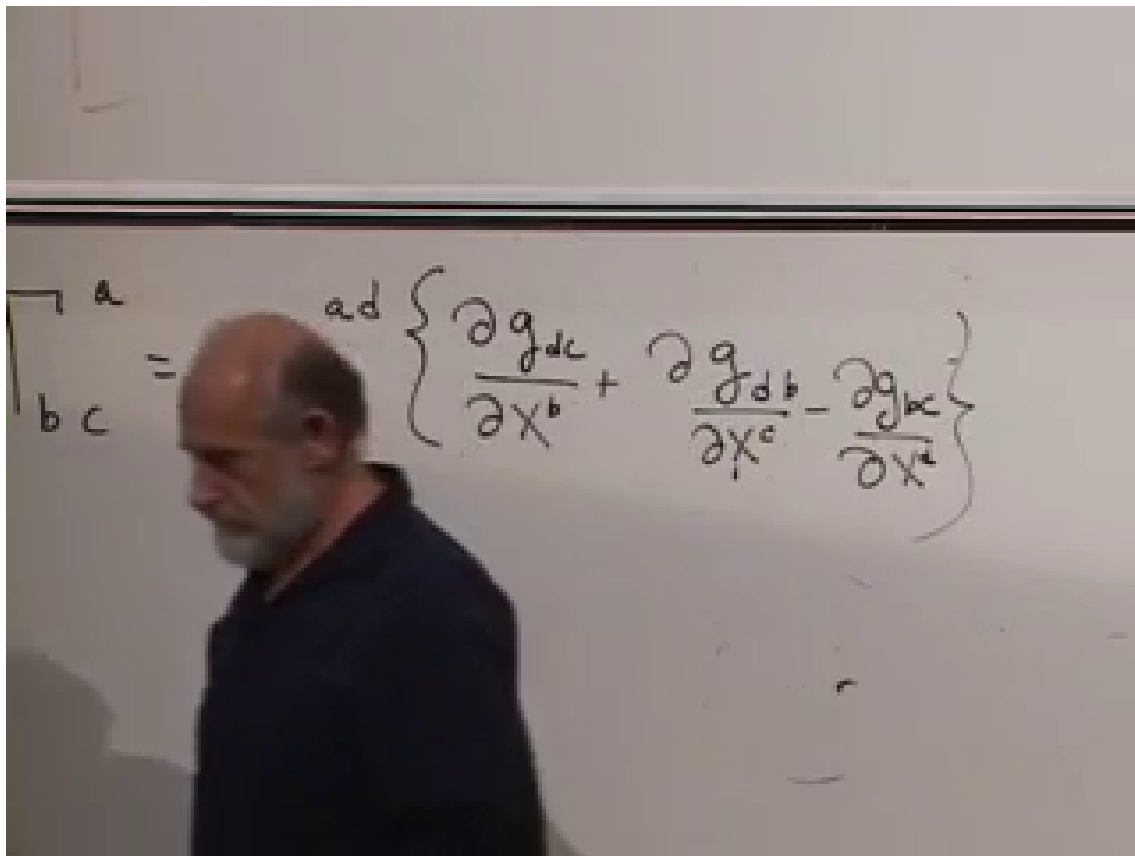


Figure 1.11: The conventional Christoffel formula. Its lower indices b, c are symmetric, and the upper index is produced by the inverse metric g^{ad} .

Lecture frame: 01:09:43.

Question from the audience. Are there two kinds of Christoffel symbol, and may their indices be raised and lowered with the metric? ^a

Response. One often writes both Γ^a_{bc} and $\Gamma_{abc} = g_{ad}\Gamma^d_{bc}$. They are related by the metric, but the connection is not a tensor, so this notation must not be mistaken for an ordinary tensor transformation law. The Levi-Civita formula makes the relationship well defined in the chosen coordinates.

^aLecture timestamp: 01:05:34–01:07:46.

Question from the audience. Should the derivatives in the Christoffel formula use x or y coordinates? ^a

Response. Use whichever coordinate system supplies the metric components. If the metric is written as $g_{mn}(y)$, differentiate with respect to y ; the resulting numbers are the connection coefficients in the y -coordinates.

^aLecture timestamp: 01:09:39–01:10:15.

Question from the audience. If all three indices are lowered, is the result symmetric in all three? ^a

Response. No. For the torsion-free Levi-Civita connection, Γ^a_{bc} is symmetric only in b and c . Lowering a does not create symmetry with that slot; the minus sign in Eq. (1.24) already distinguishes it.

^aLecture timestamp: 01:13:17–01:15:13.

The same connection acts on upper-index components with the opposite sign from lower-index components, as in Eq. (1.16).

Question from the audience. Can the metric be used to lower an index on something that is not a tensor? ^a

Response. For this metric-derived connection, the lowered coefficient is a useful and consistent abbreviation. But the result does not thereby become a tensor. Index lowering preserves tensorial character when one starts with a tensor; it cannot manufacture tensorial transformation behavior from a connection coefficient.

^aLecture timestamp: 01:15:38–01:17:23.

In D dimensions there are D^3 displayed connection coefficients: 27 in three dimensions and 64 in four. Symmetry in the two lower indices reduces the number of independent coefficients, but practical calculations can still be lengthy. The point of the construction is the logic: the connection terms convert ordinary component differentiation into a geometric operation.

1.8 What the Connection Means Locally

The fact that Γ^a_{bc} is not a tensor can be seen immediately. In flat Cartesian coordinates the metric is constant, so all Christoffel symbols vanish. In polar coordinates on the same flat plane they do not vanish. A tensor that vanished in one coordinate system would vanish in all of them.

$$\nabla_n V^m = \frac{\partial V^m}{\partial x^n} + \Gamma_{nm}^r V_r$$

$$g^{..} \partial g_{..} = \Gamma_{..}^{..}$$

Figure 1.12: Covariant differentiation of an upper-index vector. In the standard convention the connection term enters with a plus sign.

Lecture frame: 01:13:15.

The transformation law contains an inhomogeneous second-derivative term:

$$\Gamma'^a{}_{bc} = \frac{\partial y^a}{\partial x^r} \frac{\partial x^m}{\partial y^b} \frac{\partial x^n}{\partial y^c} \Gamma^r{}_{mn} + \frac{\partial y^a}{\partial x^r} \frac{\partial^2 x^r}{\partial y^b \partial y^c}. \quad (1.25)$$

That inhomogeneous term is not a defect. It is exactly what cancels the corresponding term from differentiating tensor components.

Question from the audience. Is there a coordinate-free way to understand the covariant derivative? ^a

Response. At any chosen point p , one may introduce normal coordinates for which

$$g_{mn}(p) = \delta_{mn} \quad \text{and} \quad \partial_r g_{mn}(p) = 0$$

in a Riemannian space, with the corresponding Minkowski metric in spacetime. Hence $\Gamma^a{}_{bc}(p) = 0$, and at that point the covariant derivative equals the ordinary derivative. The construction compares nearby tensors as they would be compared in the locally flattened frame. Only the statement at the chosen point is exact; curvature remains in second derivatives and prevents one normal frame from flattening an extended curved region.

^aLecture timestamp: 01:19:51–01:23:55.

Question from the audience. If the Christoffel symbols are zero, is that the definition of flat space? ^a

Response. Not at a single point. Normal coordinates can make them zero at any chosen point even in curved space. Flatness means that one can find coordinates in which they vanish throughout a neighborhood, or equivalently that the curvature tensor vanishes there. Conversely, curvilinear coordinates can make the Christoffel symbols nonzero even in flat space. In polar coordinates it is $g_{\theta\theta} = r^2$, not g_{rr} , that varies with r .

^aLecture timestamp: 01:23:56–01:25:59.

Question from the audience. Is the exterior derivative the same operation? ^a

Response. No. The exterior derivative acts on differential forms and is tensorial without a connection. In three Euclidean dimensions, the exterior derivative of a one-form corresponds, after applying the Hodge dual, to the familiar curl. That is a related but distinct construction and is deferred here.

^aLecture timestamp: 01:26:02–01:26:58.

1.9 Differentiation Along a Curve

Let a curve be described by coordinates $y^m(s)$, where s is arc length. Its tangent vector is

$$t^m = \frac{dy^m}{ds}, \quad g_{mn} t^m t^n = 1. \quad (1.26)$$

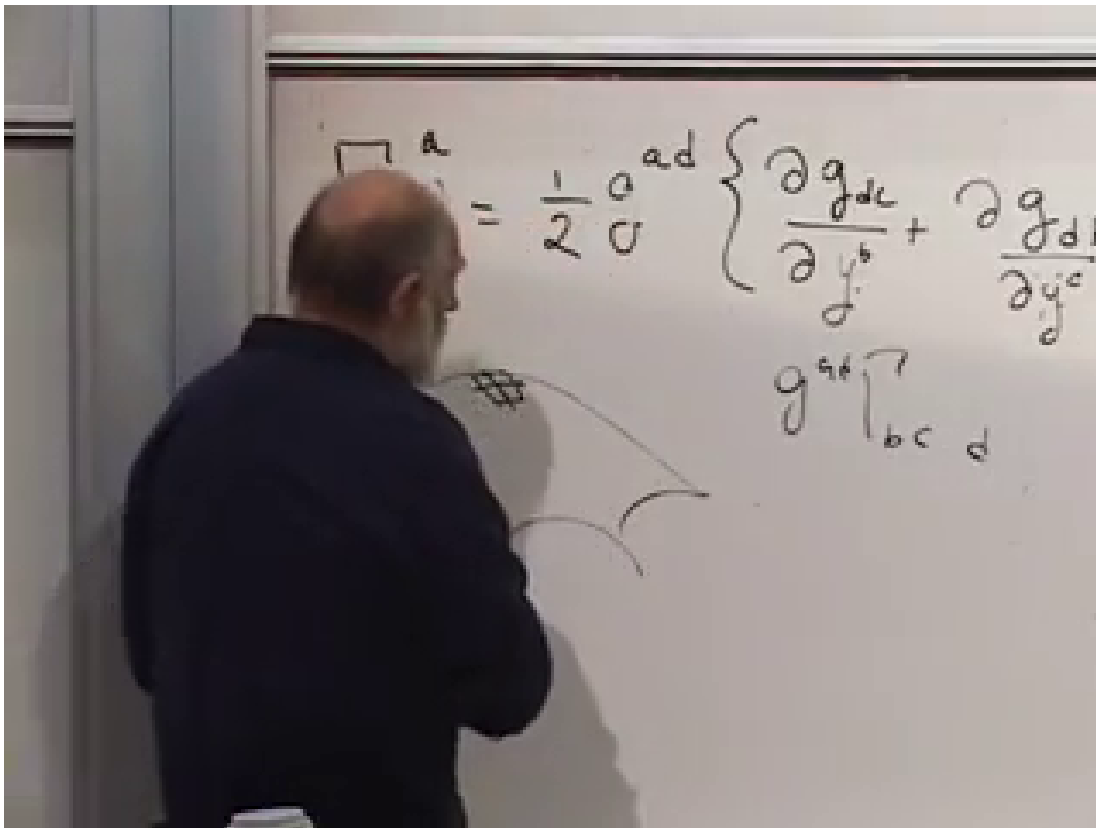


Figure 1.13: Locally Cartesian coordinates at one point of a curved surface. The metric and its first derivatives can be normalized there even when no single coordinate system flattens the whole surface.

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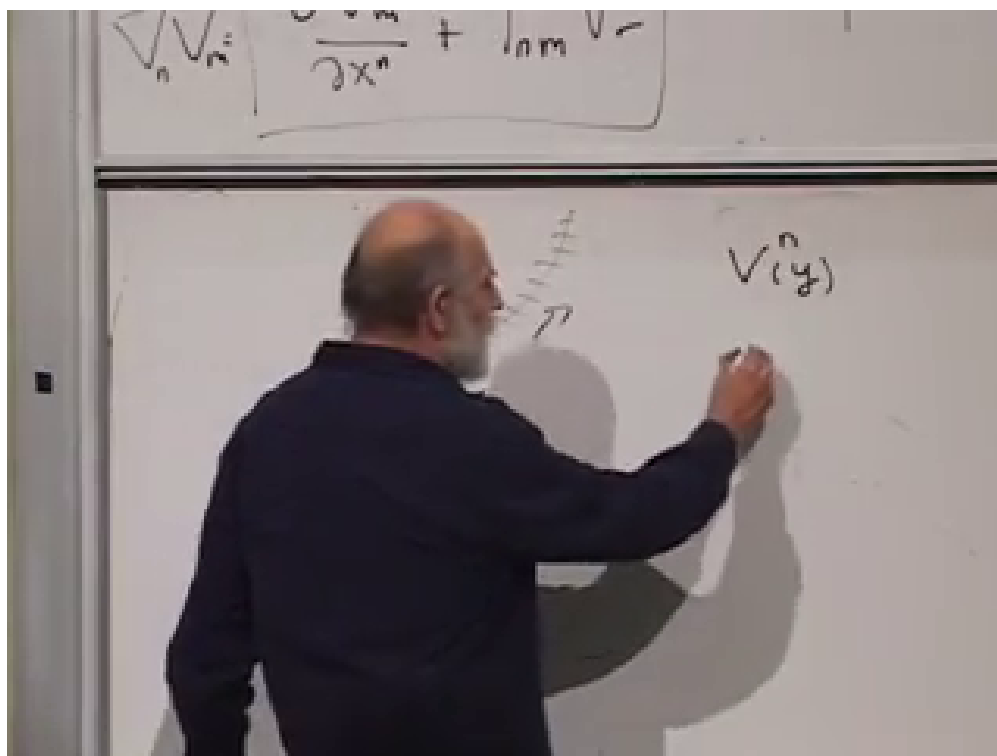


Figure 1.14: A curve parameterized by arc length s , with the tangent direction marked along the path.

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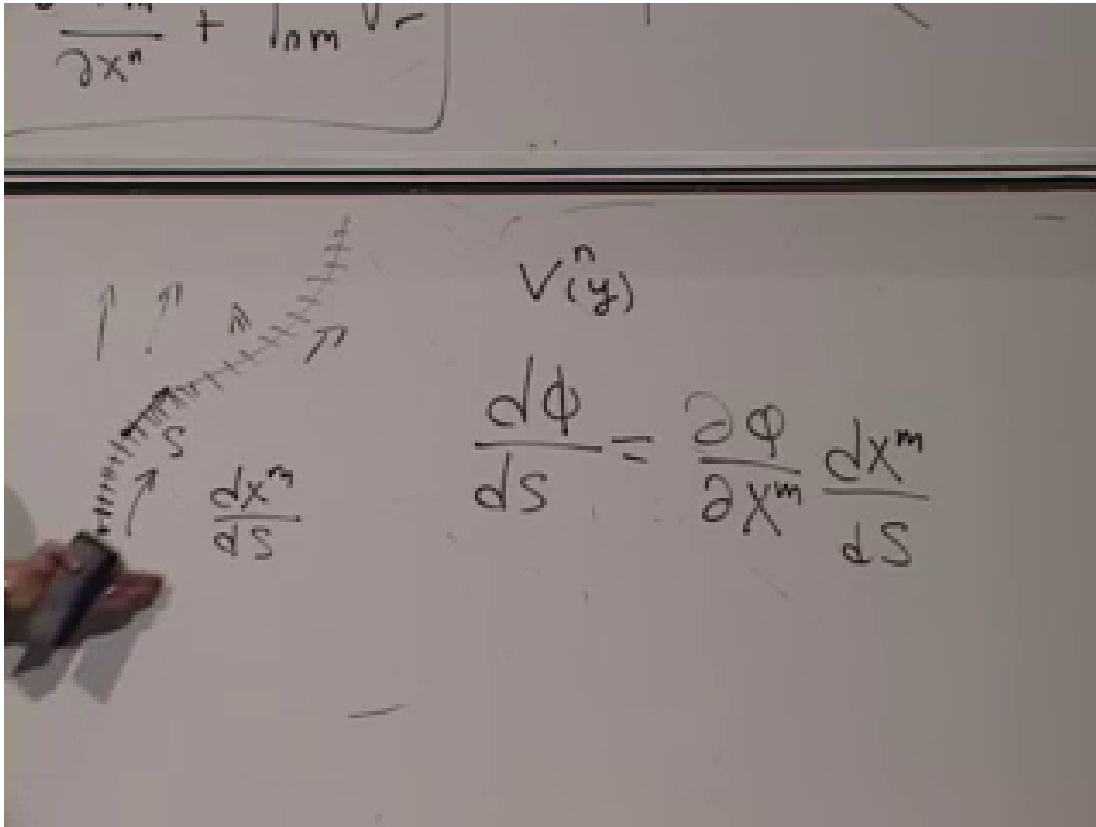


Figure 1.15: The derivative of a scalar along a curve is the contraction of its gradient with the tangent vector.

Lecture frame: 01:30:30.

For a scalar field, differentiation along the curve is an ordinary chain rule:

$$\frac{d\phi}{ds} = \frac{\partial\phi}{\partial y^m} \frac{dy^m}{ds} = t^m \partial_m \phi. \quad (1.27)$$

Question from the audience. Is dy^m/ds the slope, and why does it have unit length? ^a

Response. It is the tangent vector rather than a single coordinate slope. A slope such as dy/dx compares two coordinates; dy^m/ds lists all components of the tangent. Because s is arc length, $g_{mn}(dy^m/ds)(dy^n/ds) = 1$.

^aLecture timestamp: 01:30:33–01:31:49.

For a vector, the ordinary derivative along the curve is still not tensorial. We first take the covariant derivative and then contract its derivative index with the tangent:

$$\boxed{\frac{DV^n}{Ds} \equiv t^m \nabla_m V^n.} \quad (1.28)$$

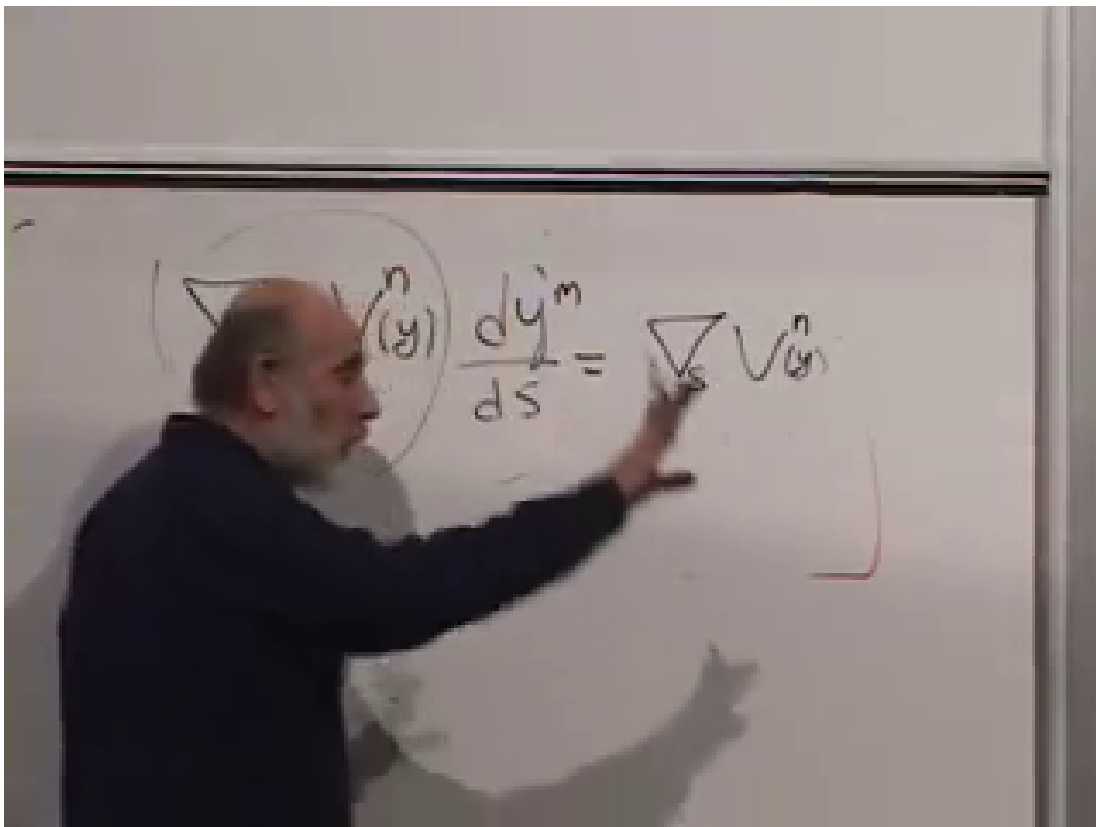


Figure 1.16: The pathwise covariant derivative is obtained by contracting the ordinary covariant derivative with the tangent direction.

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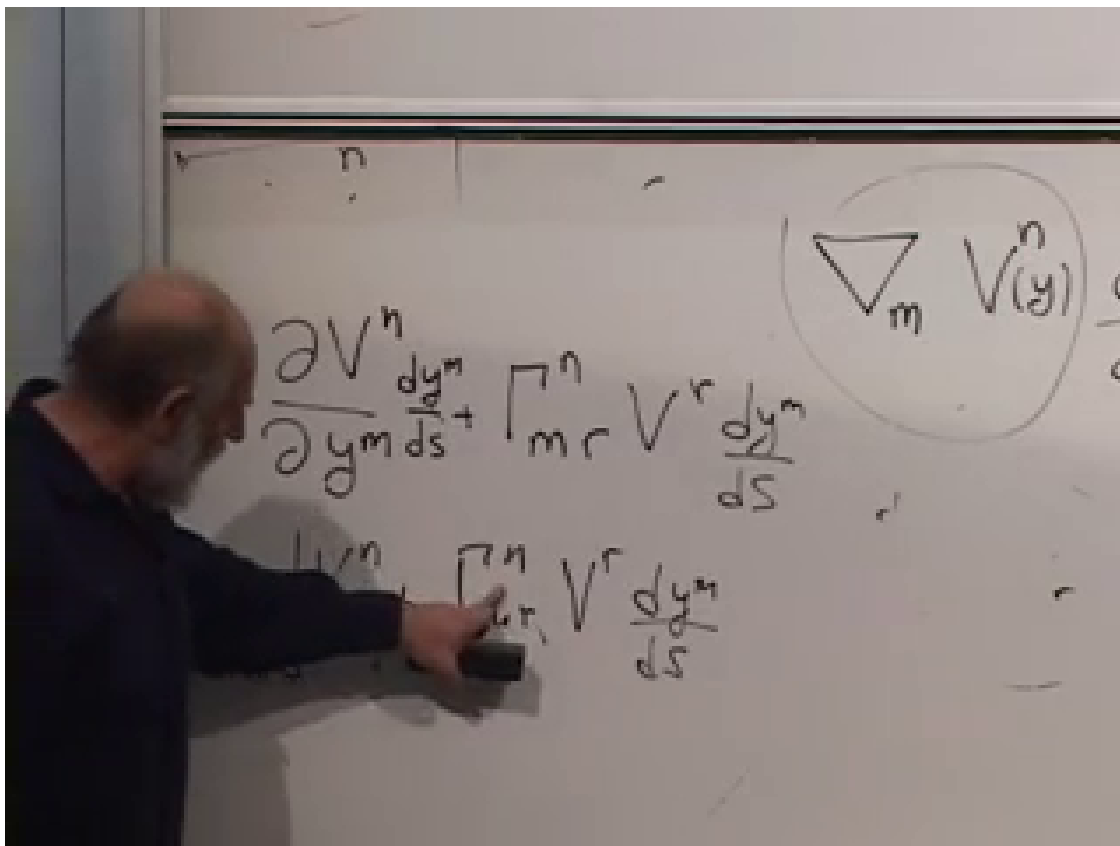


Figure 1.17: The expanded path derivative: ordinary change of the vector components plus the connection correction along the tangent.

Lecture frame: 01:39:15.

Expanding the definition,

$$\begin{aligned} \frac{DV^n}{Ds} &= (\partial_m V^n + \Gamma_{mr}^n V^r) \frac{dy^m}{ds} \\ &= \frac{dV^n}{ds} + \Gamma_{mr}^n V^r \frac{dy^m}{ds}. \end{aligned} \quad (1.29)$$

Question from the audience. Are the coordinates along this curve assumed to be Cartesian?
^a

Response. No. The point of Eq. (1.29) is that it works in arbitrary coordinates. In Cartesian coordinates on flat space the Christoffel symbols vanish and the familiar ordinary derivative is recovered.

^aLecture timestamp: 01:34:26–01:34:49.

1.10 Geodesics

Now differentiate the tangent vector itself. Set

$$V^n = t^n = \frac{dy^n}{ds}. \quad (1.30)$$

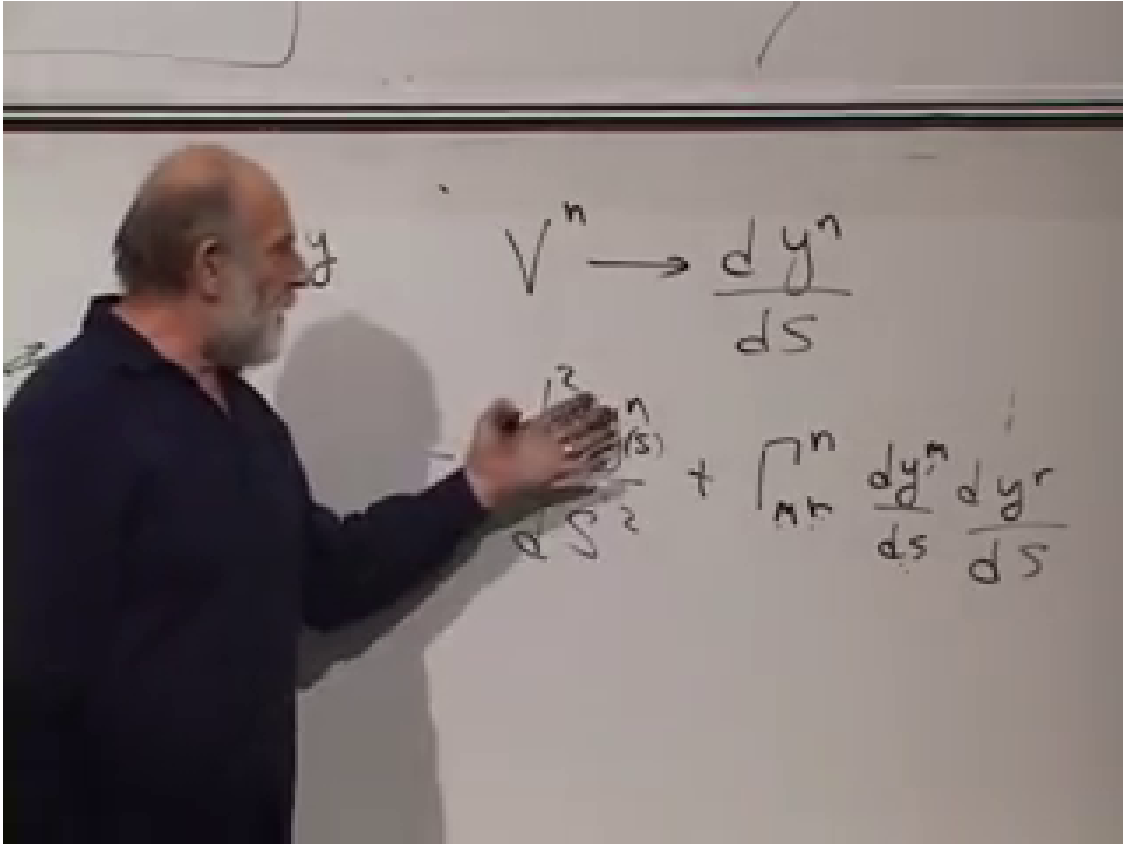


Figure 1.18: The geodesic equation on the board. A geodesic is the curve whose tangent transports itself without turning relative to the connection.

Lecture frame: 01:44:45.

Then

$$\frac{Dt^n}{Ds} = \frac{d^2 y^n}{ds^2} + \Gamma^n_{mr} \frac{dy^m}{ds} \frac{dy^r}{ds}. \quad (1.31)$$

In flat Cartesian coordinates, this vanishes along a straight line because the tangent is constant. The tensorial form lets us use the same criterion in curvilinear coordinates and in genuinely curved spaces.

Definition 1.1. A geodesic is an affinely parameterized curve whose tangent vector is covariantly constant:

$$\boxed{\frac{Dt^n}{Ds} = 0.} \quad (1.32)$$

Equivalently,

$$\boxed{\frac{d^2 y^n}{ds^2} + \Gamma^n_{mr} \frac{dy^m}{ds} \frac{dy^r}{ds} = 0.} \quad (1.33)$$

In flat space the geodesics are straight lines. In a curved space they are the straightest available curves: their tangent vectors are parallel transported along themselves. This condition can hold even though no coordinate system makes the whole space flat.

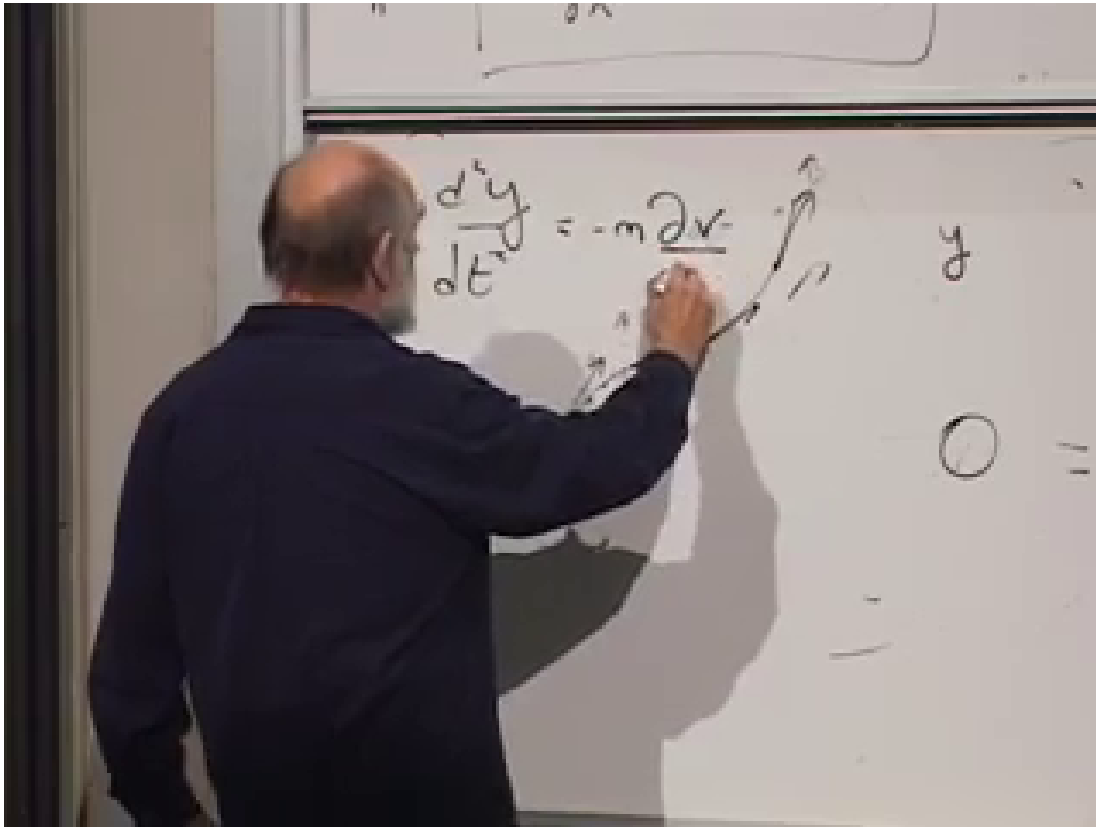


Figure 1.19: The closing comparison with Newtonian motion. Coordinate acceleration is related to a spatial derivative of the gravitational potential, while the relativistic geodesic equation contains first derivatives of the metric through Γ .

Lecture frame: 01:50:15.

Question from the audience. Could one use a contravariant derivative instead of the covariant derivative in the definition? ^a

Response. The terminology is causing the confusion. *Covariant derivative* names the coordinate-covariant operation; here it is acting on the contravariant tangent t^n . Lowering the tangent with the metric gives the equivalent lower-index equation because the Levi-Civita connection satisfies $\nabla g = 0$.

^aLecture timestamp: 01:46:19–01:47:11.

1.10.1 The road to gravity

In special relativity, a force-free particle follows a straight worldline in flat spacetime. In general relativity, a freely falling particle follows a spacetime geodesic. With proper time τ as affine parameter,

$$\frac{d^2 x^\mu}{d\tau^2} = -\Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau}. \quad (1.34)$$

The left side resembles acceleration, while the right side contains first derivatives of the metric through the Christoffel symbols.

Newtonian gravity writes

$$m \frac{d^2 \mathbf{x}}{dt^2} = -m \nabla \Phi, \quad (1.35)$$

so the mass cancels. In the weak, slowly varying field limit,

$$g_{00} \simeq 1 + \frac{2\Phi}{c^2}, \quad \Gamma^i_{00} \simeq \frac{1}{c^2} \partial_i \Phi, \quad (1.36)$$

and the spatial part of Eq. (1.34) reduces to

$$\frac{d^2 x^i}{dt^2} \simeq -\partial_i \Phi. \quad (1.37)$$

This is only a preview of the derivation, but it makes the destination clear: what Newtonian language calls gravitational force is encoded in the spacetime metric and its connection. A freely falling particle feels no nongravitational proper force; its worldline is straight in the geometric sense expressed by the geodesic equation.

One concept remains before curvature can be defined operationally: parallel transport around different paths. Curvature will measure the obstruction to making those comparisons path independent, and therefore the obstruction to flattening the geometry beyond a single point.